

GOA: from swarm to measured evidence

Grasshopper Optimisation Algorithm

ทฤษฎี • ภาพการเคลื่อนที่ • ผลทดลองที่ทำซ้ำได้

Can social interactions find a good minimum?

Compare GOA with real-number GA and parallel RLS.

Two functions • 180 independent measured runs

12-minute talk | 2D visualization | All raw runs retained

The GOA update rule

Social interaction

$$s(r) = 0.5 \exp(-r/1.5) - \exp(-r)$$

$$\mathbf{x}_i^d(t) = \mathbf{T}^d(t-1) + c_t \sum_{j \neq i} [c_t (\text{ub}^d - \text{lb}^d)/2 \cdot s(r_{ij}^d) \cdot \mathbf{u}_{ij}^d]$$

$$c_t = 1 - t (1 - 10^{-5}) / 100; \quad t = 1, \dots, 100$$

$\mathbf{T}$ = best-so-far archive; $\mathbf{u}$ = unit direction from i to j .

Eq. 2.7/2.8 of Saremi et al. (2017), with a documented normalization choice.

Exploration → exploitation

Early: explore

Higher $c \rightarrow$ stronger interactions
Repulsion spreads nearby agents
Attraction links distant agents

Late: exploit

Lower $c \rightarrow$ smaller movement
Search around the best archive
Clustering can still miss the optimum

$$r_{ij}^d = 1 + 3 |x_j^d - x_i^d| / (ub^d - lb^d) \in [1, 4]$$

This coordinate/bound mapping is our explicit implementation choice.

Synchronous updates · clip to bounds · both c factors retained · no claim of exact paper replication

Where GOA can be applied

Structural design: minimize weight subject to stress limits

Design variables → cross-sectional areas

Objective → material / structural weight

Constraints → stress or deflection limits

The original paper studies truss and cantilever design.

Our workshop tests are unconstrained mathematical functions;
we do not claim new engineering results.

A reproducible comparison

Sphere: $[-5, 5]^2$ **Rosenbrock: $[-2, 2]^2$**

2 dimensions · 30 agents · 100 iterations · 30 runs per method

Seeds: $20260930 + 10000 \times \text{function index} + \text{run index}$

Same initial points for all three methods within a function/run

3,030 actual objective calls per run: $30 + 100 \times 30$

GA: $p_c = 0.8$, $p_m = 0.1$; mutation $\sigma = 0.1 \times \text{bound width}$; no elitism

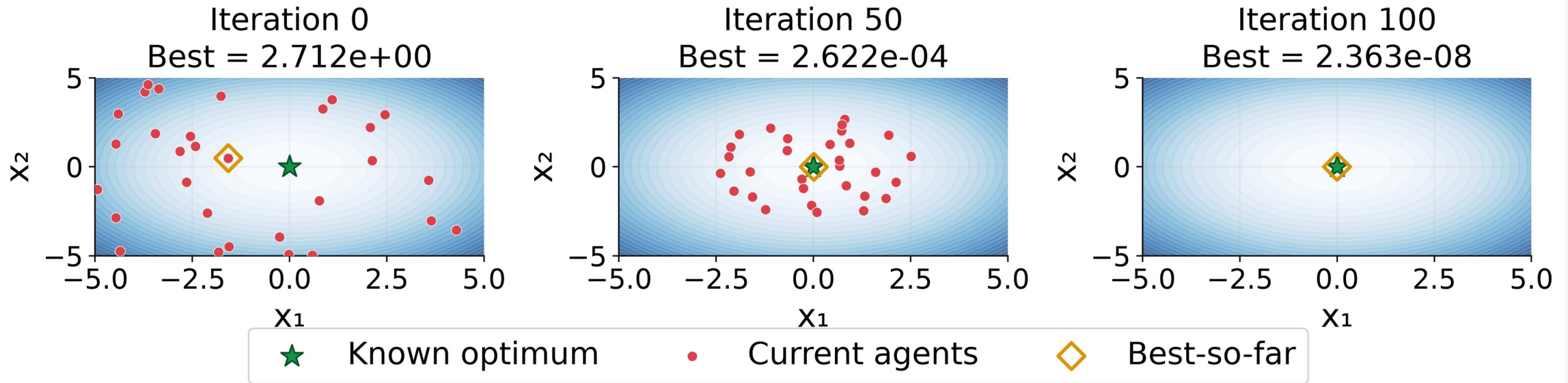
RLS: Gaussian step $\sigma = 0.3$; accept only non-worse moves

All methods archive best-so-far; time excludes graphics/export

Sphere: a smooth bowl

$$f(\mathbf{x}) = x_1^2 + x_2^2; \text{ optimum } (0, 0), f^* = 0$$

GOA · Sphere · seed 424242 · single run · color: raw f



Single illustration · preselected seed 424242 · final best = $2.363e-08$

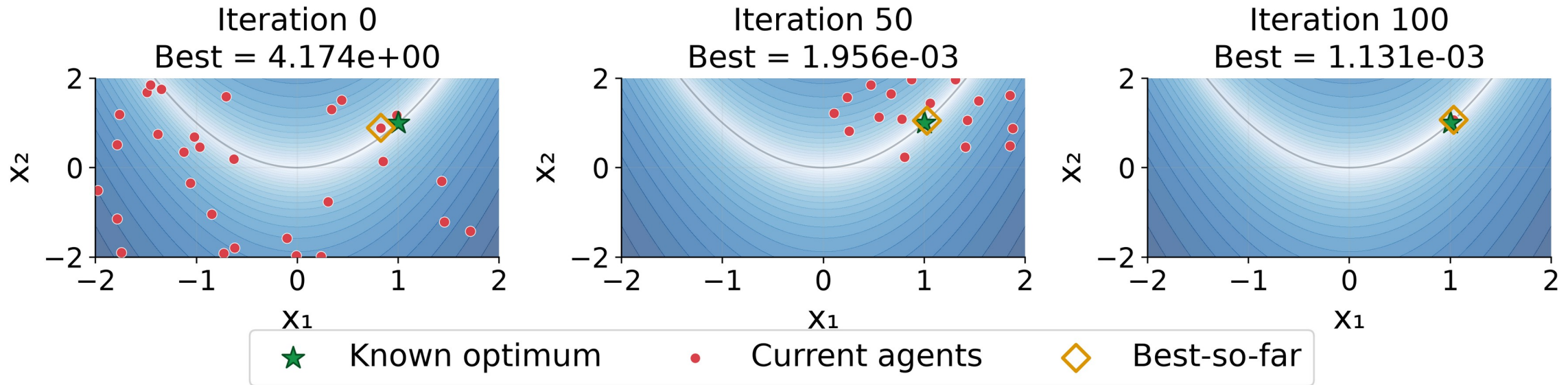
Green star = known optimum; orange diamond = archive; red points = current agents

[► Open GIF / MP4 \(16.8 s\)](#)

Rosenbrock: a narrow curved valley

$$f(\mathbf{x}) = 100(x_2 - x_1^2)^2 + (x_1 - 1)^2; \text{ optimum } (1, 1), f^* = 0$$

GOA · Rosenbrock · seed 424242 · single run · color: $\log_{10}(1 + f)$



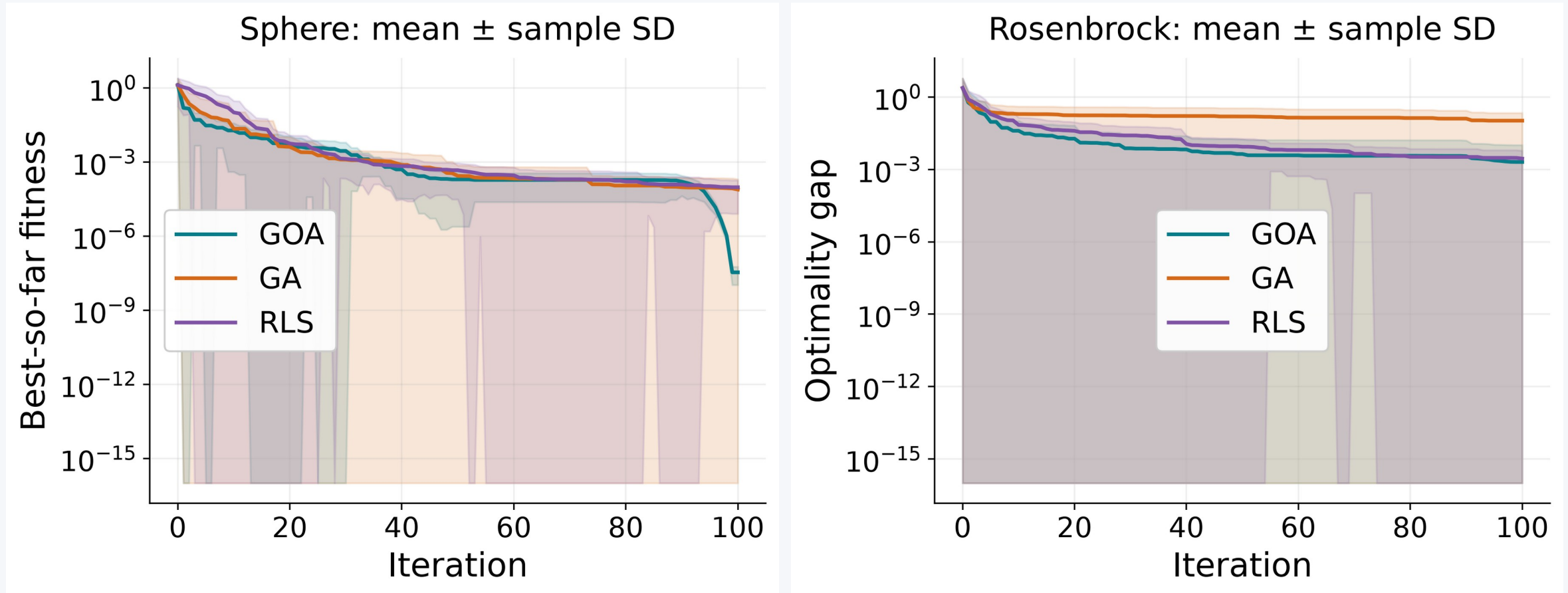
Contour: $\log_{10}(1 + f)$

Single illustration · preselected seed 424242 · final best = $1.131e-03$

Green star = known optimum; orange diamond = archive; red points = current agents

[► Open GIF / MP4 \(16.8 s\)](#)

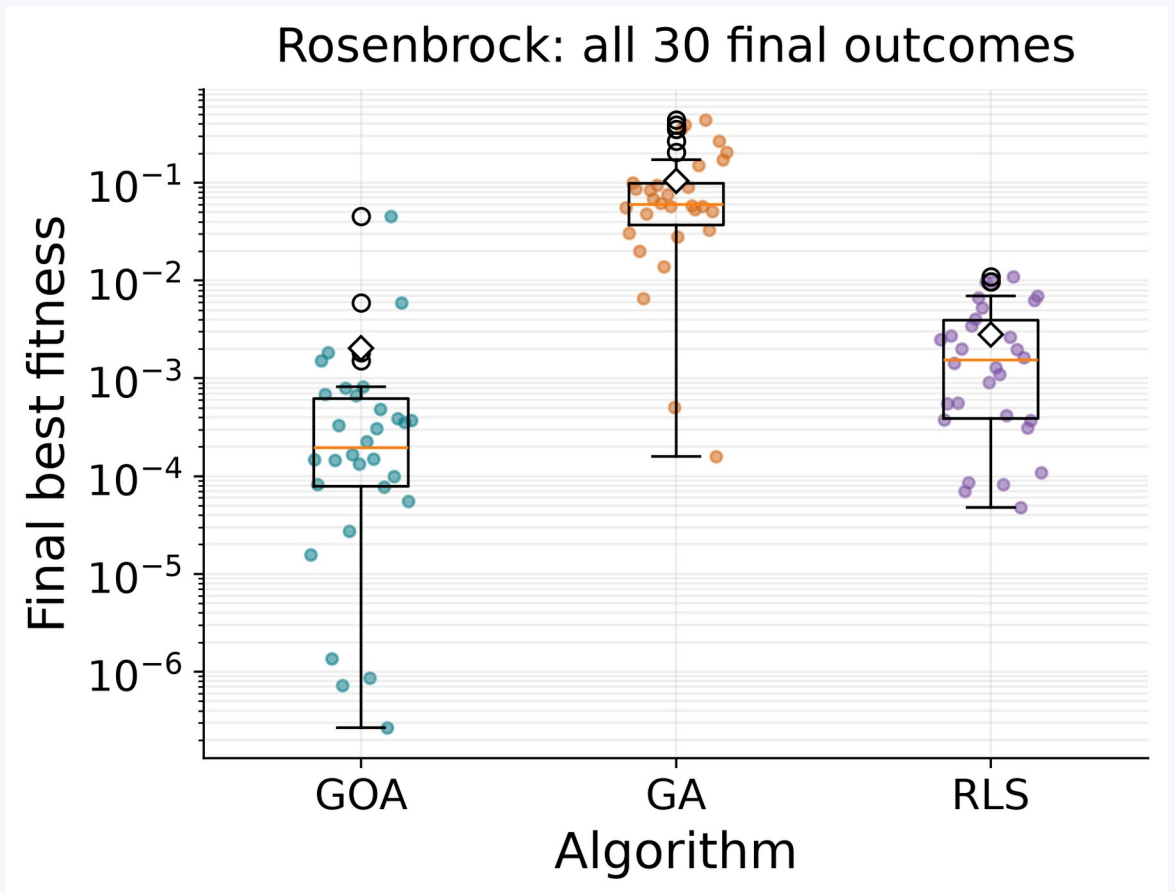
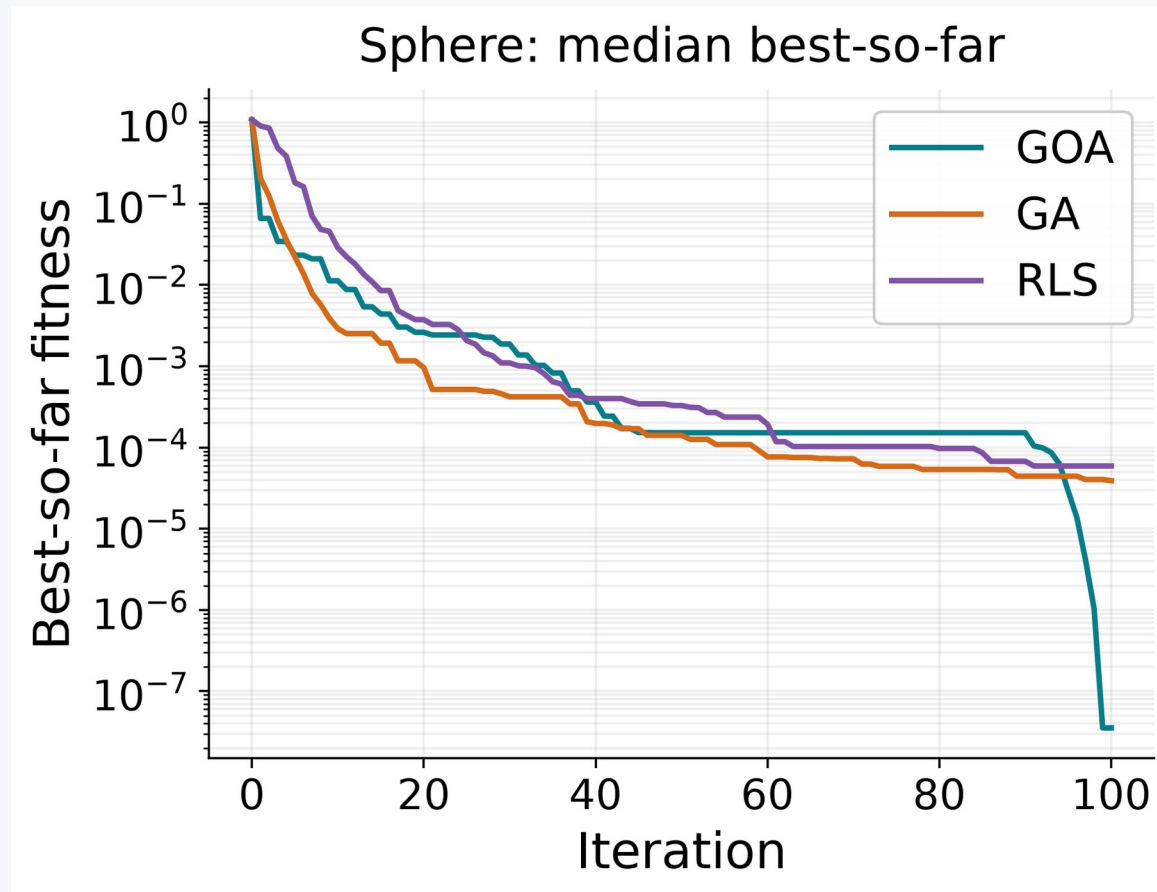
Mean convergence and optimality gap



Each line averages best-so-far over all 30 independent runs.

Shading = $\pm$ sample SD. Gap = fitness here ($f^* = 0$); log display floor = 10^{-16} .

Typical performance and run variability



GOA has the lowest median on both functions in this setup.

Rosenbrock: GOA's long upper tail means some runs still stall.

What the 180 runs show

Final fitness ↓ · all 30 runs per method · lower is better

Function / method	Mean ± SD	Median	Best	Worst	Time (s)
Sphere / GOA	3.43e-08 ± 2.37e-08	3.54e-08	8.29e-11	9.56e-08	0.0161
Sphere / GA	7.70e-05 ± 1.26e-04	3.90e-05	6.57e-07	6.71e-04	0.0165
Sphere / RLS	9.55e-05 ± 8.75e-05	5.92e-05	1.18e-06	2.97e-04	0.0094
Rosenbrock / GOA	2.04e-03 ± 8.24e-03	1.96e-04	2.66e-07	4.53e-02	0.0234
Rosenbrock / GA	1.05e-01 ± 1.15e-01	6.00e-02	1.58e-04	4.40e-01	0.0240
Rosenbrock / RLS	2.81e-03 ± 3.22e-03	1.53e-03	4.79e-05	1.10e-02	0.0161

Equal measured budgets: 3,030 calls/run. Time = mean search time on this machine.

RLS is faster here; GOA has the lowest median, but Rosenbrock worst-case GOA > RLS.

Limits and conclusions

GOA performs well here; there is no universal winner.

Only two smooth 2D functions; no real design constraints tested

Fixed parameters; no tuning study or significance claim

GOA interactions: $O(T N^2 D)$; may stagnate near an imperfect target

Swarm clustering $\neq$ proof that the known optimum was found

Next: higher dimensions, multimodal functions, sensitivity studies,
and constrained applications with explicit feasibility checks.

References & reproducibility

Saremi, Mirjalili & Lewis (2017). Grasshopper Optimisation Algorithm: Theory and application. *Advances in Engineering Software*, 105, 30–47. DOI: 10.1016/j.advengsoft.2017.01.004

Course material supplied in repository: [workshop_objective_function.pdf](#)

GA lecture: [ScienceDirect/Algorithm_ws/l9-11.pdf](#)

Our RLS baseline: [algorithms/core.py](#) (parallel Gaussian local searches)

Raw runs + config: [results/workshop/](#) | Method audit: [docs/methodology.md](#)

Rebuild: `python -m workshop all` | Thai script + Q&A: [presentation/](#)